

BRAC University
CSE460, Fall'25
Quiz 4, Set A

Name:

ID:

1. Suppose you are designing a **3 input NOR** Gate such that, the **worst-case** effective **rise resistance** is **3 times** the resistance of a unit inverter, and the **worst-case** effective **fall resistance** is equal to that of a unit inverter. The material used for making this gate is GaN for which $\mu_n = 3\mu_p$.

- (i) Draw the equivalent RC circuit considering a fanout of h.
- (ii) Find t_{pdr} , t_{cdr} , t_{pdf} , t_{cdf} considering the gate is driving 5 unit inverters (Material GaN).
- (iii) If $\lambda=10$ nm, $R=1$ k Ω , $C=12$ fF/ μ m, determine the values of the delays calculated in (ii)

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1. Suppose you are designing a **3 input NOR** Gate such that, the **worst-case** effective **rise resistance** is equal to the resistance of a unit inverter, and the **worst-case** effective **fall resistance** is **3 times** that of a unit inverter. The material used for making this gate is GaN for which $\mu_n = 3\mu_p$.

(i) Draw the equivalent RC circuit considering a fanout of h.

(ii) Find t_{pdr} , t_{cdr} , t_{pdf} , t_{cdf} considering the gate is driving 5 unit inverters.

(iii) If $\lambda=10$ nm, $R=1$ k Ω , $C=12$ fF/ μ m, determine the values of the delays calculated in (ii)